

**CIS 21031 – PRACTICAL FOR PLATFORM TECHNOLOGIES DEPARTMENT OF
ICT
FACULTY OF TECHNOLOGY
SOUTH EASTERN UNIVERSITY**

Continuous Assessment-02

Index No :- SEU/IS/20/ICT/084

Question 1

```
#!/bin/bash

addition() {
    echo "$((num1 + num2))"
}

subtraction() {
    echo "$((num1 - num2))"
}

multiplication() {
    echo "$((num1 * num2))"
}

division() {
    if [ $num2 -ne 0 ]; then
        echo "$((num1 / num2))"
    else
        echo "Error: Division by zero!"
    fi
}

modulus() {
    if [ $num2 -ne 0 ]; then
        echo "$((num1 % num2))"
    fi
}
```

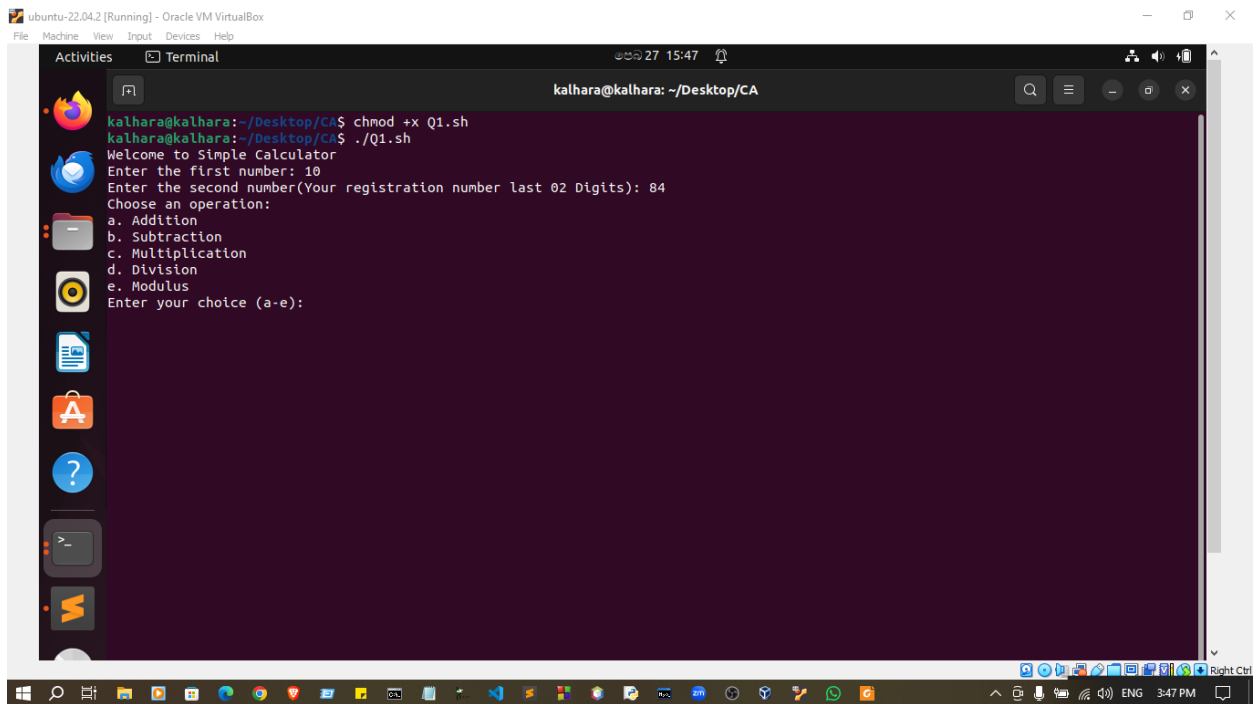
```

else
    echo "Error: Modulus by zero!"
fi
}

echo "Welcome to Simple Calculator"
read -p "Enter the first number: " num1
read -p "Enter the second number(Your registration number last 02 Digits): " num2
echo "Choose an operation:"
echo "a. Addition"
echo "b. Subtraction"
echo "c. Multiplication"
echo "d. Division"
echo "e. Modulus"
read -p "Enter your choice (a-e): " choice
case $choice in
    a) result=$((addition));;
    b) result=$((subtraction));;
    c) result=$((multiplication));;
    d) result=$((division));;
    e) result=$((modulus));;
    *) echo "Invalid choice"; exit 1;;
esac
echo "Result: $result"

```

Output:



The screenshot shows a terminal window titled 'kalhara@kalhara: ~/Desktop/CA'. The user has executed the following commands: `chmod +x Q1.sh` and `./Q1.sh`. The script output is as follows:

```
kalhara@kalhara:~/Desktop/CA$ chmod +x Q1.sh
kalhara@kalhara:~/Desktop/CA$ ./Q1.sh
Welcome to Simple Calculator
Enter the first number: 10
Enter the second number(Your registration number last 02 Digits): 84
Choose an operation:
a. Addition
b. Subtraction
c. Multiplication
d. Division
e. Modulus
Enter your choice (a-e):
```

Question 2

```
#!/bin/bash
```

```
function recursiveFunction {
    local num=$1
    if [ $num -gt 10 ]; then
        return
    fi
    echo $num
    num=$((num + 1))
    recursiveFunction $num
}
```

```
function fibonacci {
    local num=$1
    if [ $num -le 1 ]; then
```

```

    echo $num
else
    echo $(( $(fibonacci $(( $num - 1))) + $(fibonacci $(( $num - 2))) ))
fi
}

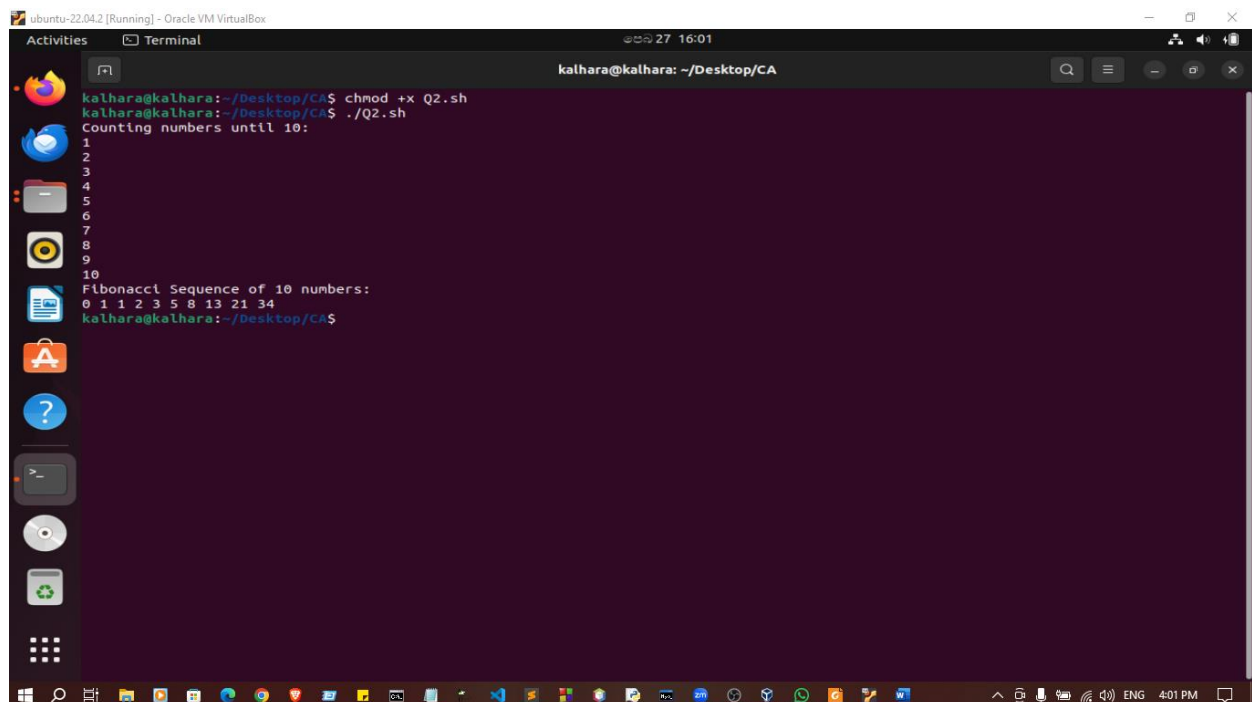
echo "Counting numbers until 10:"
recursiveFunction 1

echo "Fibonacci Sequence of 10 numbers:"
for ((i=0; i<10; i++)); do
    echo -n "$(fibonacci $i) "
done

echo

```

Output:



```

kalhara@kalhara: ~/Desktop/CA
kalhara@kalhara:~/Desktop/CA$ chmod +x Q2.sh
kalhara@kalhara:~/Desktop/CA$ ./Q2.sh
Counting numbers until 10:
1
2
3
4
5
6
7
8
9
10
Fibonacci Sequence of 10 numbers:
0 1 1 2 3 5 8 13 21 34
kalhara@kalhara:~/Desktop/CA$

```